Residual Estimator and Reliability

Element and facet residuals, local indicators, and an energy-error bound

Lecture outline

We begin with the elementwise integration-by-parts formula obtained in the previous lecture. The aim is to obtain a computable bound for the energy error.

1. Regroup element-boundary terms into interior, Neumann, and Dirichlet facet contributions.
2. Compute the residuals on a two-element 1D mesh.
3. Construct a local averaging operator and use its approximation estimates to obtain the mesh-size weights.
4. Evaluate the estimator on a two-triangle mesh and prove the global energy-error bound.

The one-dimensional example shows the effect of element size. The two-dimensional example shows why facet residuals must also be included.

Review: diffusion problem and test spaces

Recall the diffusion problem

$$\begin{aligned} -\nabla\cdot(k\nabla u) &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \Gamma_D, \\ k\nabla u\cdot \boldsymbol{n} &= q \quad \text{on } \Gamma_N, \end{aligned}$$

Here q is the **prescribed normal flux** on Γ_N, and $\boldsymbol{n}$ is the outward unit normal:

$$\boldsymbol{q} = k\nabla u, \qquad q = \boldsymbol{q}\cdot\boldsymbol{n} = k\nabla u\cdot\boldsymbol{n}.$$

The test spaces are

$$\mathcal{V}_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}, \qquad \mathcal{V}_{h,0} \subset \mathcal{V}_0.$$

Here $\mathcal{V}_{h,0}$ is the continuous Lagrange space with zero values on Γ_D. Assume $|\Gamma_D| > 0$ and $0 < k_0 \leq k \leq k_1$ throughout.

Review: variational forms and energy norm

For $f \in L^2(\Omega)$ and $q \in L^2(\Gamma_N)$, define

$$a(w, v) = \int_\Omega k \nabla w \cdot \nabla v \, \mathrm{d}x, \qquad \ell(v) = \int_\Omega f v \, \mathrm{d}x + \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The continuous and discrete equations are

$$a(u, v) = \ell(v) \quad \forall v \in \mathcal{V}_0, \qquad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.$$

Since a is symmetric and coercive on $\mathcal{V}_0$, its energy norm is

$$\|v\|_a = \sqrt{a(v, v)} = \left(\int_\Omega k |\nabla v|^2 \, \mathrm{d}x \right)^{1/2}.$$

The same a and ℓ define both problems. The test space changes from $\mathcal{V}_0$ to its finite element subspace $\mathcal{V}_{h,0}$.

Review: residual and energy error

Since $u, u_h \in \mathcal{V}_0$, the error $e = u - u_h$ also belongs to $\mathcal{V}_0$. For any $v \in \mathcal{V}_0$, the continuous equation gives $\ell(v) = a(u, v)$. Starting from the definition of the residual,

$$\begin{aligned} R(u_h; v) &= \ell(v) - a(u_h, v) \\ &= a(u, v) - a(u_h, v) \qquad \text{(continuous equation)} \\ &= a(u - u_h, v) = a(e, v) \quad \text{(bilinearity)}. \end{aligned}$$

We may take $v = e$ because $e \in \mathcal{V}_0$. This gives

$$\boxed{R(u_h; e) = a(e, e) = \|e\|_a^2 .}$$

The last equality uses the definition $\|e\|_a = \sqrt{a(e, e)}$. Thus testing the residual with the error gives the squared energy error.

Review: integration by parts on each element

Start with the product rule on $\mathcal{K}_e$:

$$\nabla\cdot(vk\nabla u_h) = k\nabla u_h\cdot\nabla v + v\nabla\cdot(k\nabla u_h).$$

Integrate over $\mathcal{K}_e$ and apply the divergence theorem:

$$\int_{\partial\mathcal{K}_e} v(k\nabla u_h\cdot\boldsymbol{n}_e)\,\mathrm{d}s = \int_{\mathcal{K}_e} k\nabla u_h\cdot\nabla v\,\mathrm{d}x + \int_{\mathcal{K}_e} v\nabla\cdot(k\nabla u_h)\,\mathrm{d}x.$$

Rearrange to isolate the first volume integral:

$$\begin{aligned}\int_{\mathcal{K}_e} k\nabla u_h\cdot\nabla v\,\mathrm{d}x = &-\int_{\mathcal{K}_e}\nabla\cdot(k\nabla u_h)v\,\mathrm{d}x\\ &+\int_{\partial\mathcal{K}_e}(k\nabla u_h\cdot\boldsymbol{n}_e)v\,\mathrm{d}s.\end{aligned}$$

Here $\boldsymbol{n}_e$ is the outward unit normal of $\mathcal{K}_e$.

Expanding the residual before integration by parts

Start from $R(u_h; v) = \ell(v) - a(u_h, v)$. Substitute the definitions of ℓ and a:

$$R(u_h; v) = \int_\Omega fv \, \mathrm{d}x + \int_{\Gamma_N} qv \, \mathrm{d}s - \int_\Omega k \nabla u_h \cdot \nabla v \, \mathrm{d}x.$$

The elements partition Ω, so each volume integral is a sum of element integrals:

$$\begin{aligned} R(u_h; v) = &\sum_e \int_{\mathcal{K}_e} fv \, \mathrm{d}x + \int_{\Gamma_N} qv \, \mathrm{d}s \\ &- \sum_e \int_{\mathcal{K}_e} k \nabla u_h \cdot \nabla v \, \mathrm{d}x. \end{aligned}$$

Apply the preceding Green's formula to each integral in the second line. The minus sign in front of this sum changes both signs in that formula.

Element and boundary contributions to the residual

In the last line of the preceding slide, substitute Green's formula:

$$R(u_h; v) = \sum_e \int_{\mathcal{K}_e} \big(f + \nabla\cdot(k\nabla u_h)\big) v \, \mathrm{d}x \\ - \sum_e \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e) v \, \mathrm{d}s + \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The first line is already separated by element. In the second line:

- an interior facet appears twice, once from each neighboring element;
- a Neumann facet appears once and combines with the prescribed-flux integral; and
- a Dirichlet facet contributes zero because $v = 0$ on Γ_D.

Interior facets: combining the two boundary terms

Let $\mathcal{F} = \partial\mathcal{K}^+ \cap \partial\mathcal{K}^-$. The outward normals are $\boldsymbol{n}_\mathcal{F}$ for $\mathcal{K}^+$ and $-\boldsymbol{n}_\mathcal{F}$ for $\mathcal{K}^-$. The two boundary terms are

$$
\begin{aligned}
&- \int_\mathcal{F} k^+ \nabla u_h^+ \cdot \boldsymbol{n}_\mathcal{F}\, v \, \mathrm{d}s - \int_\mathcal{F} k^- \nabla u_h^- \cdot (-\boldsymbol{n}_\mathcal{F})\, v \, \mathrm{d}s \\
&\qquad = - \int_\mathcal{F} \left(k^+ \nabla u_h^+ - k^- \nabla u_h^-\right) \cdot \boldsymbol{n}_\mathcal{F}\, v \, \mathrm{d}s.
\end{aligned}
$$

The superscripts identify values from the two elements. The same v appears in both terms because its values match across the shared facet. Define

$$
j_\mathcal{F} = -\left(k^+ \nabla u_h^+ - k^- \nabla u_h^-\right) \cdot \boldsymbol{n}_\mathcal{F}.
$$

If the element labels are exchanged, the new normal is $-\boldsymbol{n}_\mathcal{F}$. The new residual is

$$
-(k^- \nabla u_h^- - k^+ \nabla u_h^+) \cdot (-\boldsymbol{n}_\mathcal{F}) = j_\mathcal{F}.
$$

The flux difference and the normal both change sign, so their dot product is unchanged.

Boundary facets: Neumann mismatch and Dirichlet values

On a Neumann facet, combine the two boundary terms from the residual:

$$-\int_{\mathcal{F}}(k\nabla u_h\cdot\boldsymbol{n})v\ \mathrm{d}s+\int_{\mathcal{F}}qv\ \mathrm{d}s=\int_{\mathcal{F}}(q-k\nabla u_h\cdot\boldsymbol{n})v\ \mathrm{d}s.$$

The Neumann-facet residual is therefore

$$j_{\mathcal{F}}=\underbrace{q}_{\text{prescribed}}\ \underbrace{-k\nabla u_h\cdot\boldsymbol{n}}_{\text{computed}}.$$

On a Dirichlet facet, the test function satisfies $v=0$, so

$$-\int_{\mathcal{F}}(k\nabla u_h\cdot\boldsymbol{n})v\ \mathrm{d}s=0.$$

There is no Dirichlet-facet term in the residual representation.

Complete element-and-facet residual representation

Let $\mathcal{F}_h^{\mathrm{int}}$ denote interior facets and $\mathcal{F}_h^N$ boundary facets on Γ_N. The residuals are

$$r_e = f + \nabla\cdot(k\nabla u_h) \quad \text{on } \mathcal{K}_e.$$

$$j_{\mathcal{F}} = \begin{cases} -(k^+\nabla u_h^+ - k^-\nabla u_h^-)\cdot \boldsymbol{n}_{\mathcal{F}}, & \mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}, \\ q - k\nabla u_h\cdot\boldsymbol{n}, & \mathcal{F}\in\mathcal{F}_h^N. \end{cases}$$

The residual is therefore

$$\boxed{R(u_h; v) = \sum_e \int_{\mathcal{K}_e} r_e v \,\mathrm{d}x + \sum_{\mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}\cup\mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} v \,\mathrm{d}s.}$$

Each interior facet appears once in this exact identity.

What the two residuals measure

Element residual
$r_e = f + \nabla \cdot (k \nabla u_h)$
balance defect inside $\mathcal{K}_e$

Facet residual
$j_{\mathcal{F}}$
flux jump or mismatch

For P_1 elements and elementwise-constant k, ∇u_h and $k \nabla u_h$ are constant on each element. Hence

$$\nabla \cdot (k \nabla u_h) = 0, \qquad r_e = f|_{\mathcal{K}_e}.$$

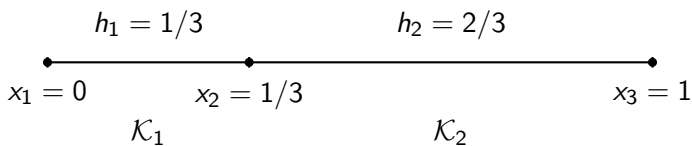
The element residual then reduces to the prescribed source. The computed-gradient information is contained in the facet residuals.

Nonuniform 1D example: mesh and computed solution

Consider the problem

$$-u'' = 1 \quad \text{on } (0,1), \qquad u(0) = 0, \qquad u'(1) = 0.$$

Here $k = 1$, $f = 1$, and $q = 0$ at $x = 1$. Take the computed nodal values $U_1 = 0$, $U_2 = 5/18$, and $U_3 = 1/2$ as given.



$$u_h'|_{\mathcal{K}_1} = \frac{U_2 - U_1}{h_1} = \frac{5/18 - 0}{1/3} = \frac{5}{6}, \qquad u_h'|_{\mathcal{K}_2} = \frac{U_3 - U_2}{h_2} = \frac{1/2 - 5/18}{2/3} = \frac{1}{3}.$$

Since $u_h'' = 0$ inside each element, $r_1 = r_2 = f + u_h'' = 1$. The remaining residuals are at the nodes.

The 1D residual includes nodal facet terms

In one dimension, facet integrals become evaluations at nodes. At x_2, the outward normals are $n_1 = +1$ from $\mathcal{K}_1$ and $n_2 = -1$ from $\mathcal{K}_2$. Substitute the computed slopes into

$$j_2 = -\big(u_h'|_{\mathcal{K}_1} n_1 + u_h'|_{\mathcal{K}_2} n_2\big) = -\left(\frac{5}{6}(+1) + \frac{1}{3}(-1)\right) = -\frac{1}{2}.$$

At $x_3 = 1$, the prescribed normal datum is $q = 0$, so

$$j_3 = q - u_h'(1)\,(+1) = 0 - \frac{1}{3} = -\frac{1}{3}.$$

Substitute $r_1 = r_2 = 1$, $j_2 = -1/2$, and $j_3 = -1/3$ into the residual representation:

$$\begin{aligned} R(u_h; v) &= \sum_{e=1}^{2} \int_{\mathcal{K}_e} r_e v \, \mathrm{d}x + j_2 v(x_2) + j_3 v(x_3) \\ &= \int_{\mathcal{K}_1} v \, \mathrm{d}x + \int_{\mathcal{K}_2} v \, \mathrm{d}x - \frac{1}{2} v(x_2) - \frac{1}{3} v(x_3). \end{aligned}$$

Here $v \in H^1(0,1)$ and $v(0) = 0$, so there is no term at $x_1 = 0$.

Constructing a residual estimator

Remove the unknown error from the residual identity, identify the mesh-size weights, and obtain a computable upper bound.

The unknown error in the residual identity

Combining $R(u_h; e) = \|e\|_a^2$ with the residual representation gives

$$\|e\|_a^2 = R(u_h; e) = \sum_{e'} \int_{\mathcal{K}_{e'}} r_{e'} e \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} e \, \mathrm{d}s.$$

Here e' indexes elements, while e is the error function. All residuals are known, but e is not. By Cauchy–Schwarz,

$$\left| \int_{\mathcal{K}_{e'}} r_{e'} e \, \mathrm{d}x \right| \leq \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \|e\|_{L^2(\mathcal{K}_{e'})}$$

which still contains the unknown local error norm.
The goal is a bound $\|e\|_a \leq C_{\mathrm{rel}} \eta$, where η uses only u_h, the data, and the mesh. A finite element approximation of e will let us introduce the local mesh sizes into this bound.

Testing the residual with the error

Both u and u_h satisfy the same zero Dirichlet condition. Hence $e = u - u_h \in \mathcal{V}_0$ is an admissible test function.
Take $v = e$ in the continuous equation $a(u, v) = \ell(v)$, giving $\ell(e) = a(u, e)$. Starting from the residual definition,

$$
\begin{aligned}
R(u_h; e) &= \ell(e) - a(u_h, e) && \text{(definition of the residual)} \\
&= a(u, e) - a(u_h, e) && \text{(continuous equation, } v = e\text{)} \\
&= a(u - u_h, e) && \text{(bilinearity in the first argument)} \\
&= a(e, e) && (e = u - u_h) \\
&= \|e\|_a^2 && \text{(definition of the energy norm).}
\end{aligned}
$$

Subtracting a finite element function

The preceding calculation gives $R(u_h; e) = \|e\|_a^2$. The discrete equation is $a(u_h, v_h) = \ell(v_h)$ for every $v_h \in \mathcal{V}_{h,0}$. It gives the separate identity

$$R(u_h; v_h) = \ell(v_h) - a(u_h, v_h) = \ell(v_h) - \ell(v_h) = 0.$$

Now use linearity of the residual in its test function:

$$\begin{aligned} R(u_h; e - v_h) &= R(u_h; e) - \underbrace{R(u_h; v_h)}_{0} \\ &= R(u_h; e) = \|e\|_a^2. \end{aligned}$$

We can choose v_h to approximate e without changing the residual value.

An approximation that does not need nodal values

For a continuous function v, the nodal interpolant from Chapter 7 is

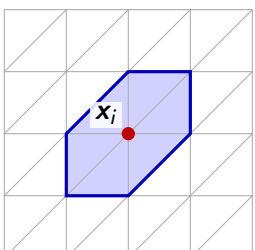
$$\mathcal{I}_h v = \sum_i v(\boldsymbol{x}_i)\phi_i.$$

But $e = u - u_h$ is known only to belong to $H^1(\Omega)$. In two and three dimensions, an H^1 function need not have well-defined point values $e(\boldsymbol{x}_i)$.
A local average is well defined even when a point value is not. We will replace $v(\boldsymbol{x}_i)$ by an average over the elements meeting at $\boldsymbol{x}_i$.
This defines a **quasi-interpolation operator** $\mathcal{Q}_h$. Its output is a continuous finite element function, so it can serve as v_h in the subtraction identity.

A vertex patch

A **vertex patch** ω_i contains all elements meeting at the vertex $\boldsymbol{x}_i$:

$$\omega_i = \bigcup_{\mathcal{K}_e:\, \boldsymbol{x}_i \in \overline{\mathcal{K}_e}} \overline{\mathcal{K}_e}.$$



The six blue triangles form ω_i in this mesh. The nodal basis function ϕ_i is nonzero only on these elements, so

$$\operatorname{supp} \phi_i = \omega_i.$$

The average used for coefficient i will use v on this blue region.

The Clément averaging construction

For P_1 elements, replace the nodal value by the **patch average**

$$c_i(v) = \frac{1}{|\omega_i|} \int_{\omega_i} v \, \mathrm{d}x, \qquad \boldsymbol{x}_i \notin \overline{\Gamma_D}.$$

Here $|\omega_i|$ is the area or volume of the patch. Set $c_i(v) = 0$ at Dirichlet vertices. With N_n mesh vertices, form

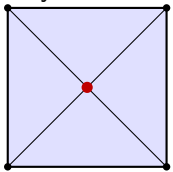
$$\boxed{\mathcal{Q}_h v = \sum_{i=1}^{N_n} c_i(v)\phi_i.}$$

The shared nodal coefficients make $\mathcal{Q}_h v$ continuous. On a Dirichlet facet, all vertex coefficients are zero, so the linear function is zero on the whole facet. Thus $\mathcal{Q}_h v \in \mathcal{V}_{h,0}$ for $v \in \mathcal{V}_0$.

Averaging version: Schöberl, §4.3, Definition 74, p. 64.

A concrete quasi-interpolant

Take $\Omega = (-1,1)^2$, $\Gamma_D = \partial\Omega$, and $v(x_1, x_2) = (1 - x_1^2)(1 - x_2^2)$. This function is zero on the boundary.



$\boldsymbol{x}_1 = (0,0)$
$\omega_1 = \overline{\Omega}$

The only free vertex is $\boldsymbol{x}_1$, with $|\omega_1| = 4$. First evaluate

$$\int_{-1}^{1} (1 - t^2)\,\mathrm{d}t = \left[t - \frac{t^3}{3}\right]_{-1}^{1} = \frac{4}{3}.$$

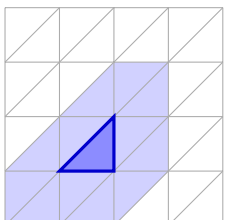
The product form of v lets us factor the double integral. All boundary coefficients are zero.

$$\begin{aligned} c_1(v) &= \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1} (1 - x_1^2)(1 - x_2^2)\,\mathrm{d}x_1\,\mathrm{d}x_2 \\ &= \frac{1}{4}\left(\int_{-1}^{1} (1 - t^2)\,\mathrm{d}t\right)^2 = \frac{1}{4}\left(\frac{4}{3}\right)^2 = \frac{4}{9}. \end{aligned}$$

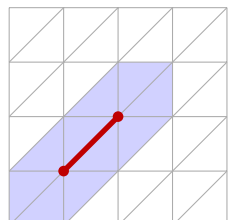
Thus $\mathcal{Q}_h v = \frac{4}{9}\phi_1$, whereas $\mathcal{I}_h v = \phi_1$ since $v(0,0) = 1$.

The patches in the local estimates

An **element patch** ω_e contains all elements sharing a vertex with $\mathcal{K}_e$. A **facet patch** $\omega_{\mathcal{F}}$ contains all elements sharing a vertex with $\mathcal{F}$.



Dark triangle: $\mathcal{K}_e$
Blue region: ω_e



Red edge: $\mathcal{F}$
Blue region: $\omega_{\mathcal{F}}$

These neighborhoods appear because each coefficient of $\mathcal{Q}_h v$ uses a vertex patch, rather than a value on one element alone.

Assumptions for the residual error bound

The estimates below use the following setting, as in Chapter 8.

- The mesh consists of affine triangles or tetrahedra. It is conforming, boundary fitted, and shape regular. The geometric constants then stay bounded as the mesh is refined.
- The finite element space is the continuous Lagrange space of fixed degree $p \geq 1$. In particular, it contains the piecewise-linear approximation $\mathcal{Q}_h v$.
- The coefficient satisfies $k|_{\mathcal{K}_e} \in C^1(\overline{\mathcal{K}_e})$. It may jump between elements. The element divergence and facet fluxes can therefore be evaluated separately on each element.
- The discrete equations are solved exactly, and the forms are evaluated exactly. These assumptions give $R(u_h; v_h) = 0$ for all $v_h \in \mathcal{V}_{h,0}$.

Local approximation estimates for $\mathcal{Q}_h$

For a shape-regular mesh in $d = 2$ or 3, use the following standard quasi-interpolation estimates for $v \in \mathcal{V}_0$:

$$\boxed{\begin{aligned} \|v - \mathcal{Q}_h v\|_{L^2(\mathcal{K}_e)} &\le C_Q h_e \, |v|_{H^1(\omega_e)}, \\ \|v - \mathcal{Q}_h v\|_{L^2(\mathcal{F})} &\le C_Q h_{\mathcal{F}}^{1/2} \, |v|_{H^1(\omega_{\mathcal{F}})}. \end{aligned}}$$

Here $h_e = \operatorname{diam}(\mathcal{K}_e)$, $h_{\mathcal{F}} = \operatorname{diam}(\mathcal{F})$, and

$$|v|_{H^1(\omega)} = \left(\int_\omega |\nabla v|^2 \, \mathrm{d}x \right)^{1/2}.$$

The constant C_Q does not depend on mesh size. We use these estimates without proving them. They are Hypothesis (Q) in Chapter 8.

Schöberl, §4.3, Theorem 75, p. 65. The element and facet powers are h_e and $h_{\mathcal{F}}^{1/2}$, respectively.

Applying the subtraction identity to the error

The operator gives $\mathcal{Q}_h e \in \mathcal{V}_{h,0}$. Set $v_h = \mathcal{Q}_h e$ in the subtraction identity:

$$\begin{aligned} \|e\|_a^2 &= R(u_h; e) \\ &= R(u_h; e) - \underbrace{R(u_h; \mathcal{Q}_h e)}_{0} \\ &= R(u_h; e - \mathcal{Q}_h e). \end{aligned}$$

Write $w = e - \mathcal{Q}_h e \in \mathcal{V}_0$. Substituting this particular test function into the element-and-facet residual representation gives

$$\boxed{\|e\|_a^2 = \sum_{e'} \int_{K_{e'}} r_{e'} w \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s.}$$

Neither e nor $\mathcal{Q}_h e$ is computed when evaluating the estimator.

The element term produces the weight h_e

Recall $w = e - \mathcal{Q}_h e$. For one element $\mathcal{K}_{e'}$,

$$\left| \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \right| \leq \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, \|w\|_{L^2(\mathcal{K}_{e'})}$$

$$\leq C_Q \qquad \underbrace{h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}}_{\text{computable weighted element residual}} \qquad |e|_{H^1(\omega_{e'})} \, .$$

Use Cauchy–Schwarz on $\mathcal{K}_{e'}$, then the element estimate with $v = e$, for which $v - \mathcal{Q}_h v = w$.
The square of the computable factor is

$$\left(h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \right)^2 = h_{e'}^2 \, \|r_{e'}\|^2_{L^2(\mathcal{K}_{e'})} \, .$$

This is the element contribution used in the estimator.

The facet term produces the weight $h_{\mathcal{F}}^{1/2}$

For one interior or Neumann facet,

$$\left| \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \right| \leq \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \, \|w\|_{L^2(\mathcal{F})}$$
$$\leq C_Q \qquad \underbrace{h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}}_{\text{computable weighted facet residual}} \qquad |e|_{H^1(\omega_{\mathcal{F}})} .$$

Use Cauchy–Schwarz on $\mathcal{F}$, then the facet estimate with $v = e$, for which $v - \mathcal{Q}_h v = w$. The square of the computable factor is

$$\left(h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \right)^2 = h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2 .$$

This is the facet contribution used in the estimator.

The global residual estimator

Let

$$\mathcal{F}_h^* = \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N.$$

For $d = 2$ or 3, define the **global residual estimator** by collecting the squared computable factors from the two bounds:

$$\boxed{\eta^2 = \sum_e h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} + \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} \cdot}$$

Every quantity in η is computable from the mesh, the data, and u_h.
The reliability proof will compare η with the unknown energy error.

Local indicators distribute the global estimator

For each element $\mathcal{K}_e$, define the **local error indicator** by

$$\eta_e^2 = h_e^2 \left\| r_e \right\|_{L^2(\mathcal{K}_e)}^2 + \frac{1}{2} \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{\mathrm{int}}}} h_{\mathcal{F}} \left\| j_{\mathcal{F}} \right\|_{L^2(\mathcal{F})}^2 + \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \in \mathcal{F}_h^{N}}} h_{\mathcal{F}} \left\| j_{\mathcal{F}} \right\|_{L^2(\mathcal{F})}^2 .$$

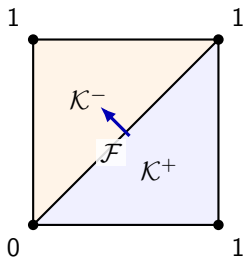
On summing over elements, each interior contribution has total weight $1/2 + 1/2 = 1$. Each Neumann contribution is counted once. Hence

$$\eta^2 = \sum_e \eta_e^2 .$$

The values η_e help select elements for refinement. They are not exact local errors.

Two-triangle example: zero element residuals

Consider $-\Delta u = 0$ on $\Omega = (0,1)^2$, with Dirichlet values shown at the four vertices and varying linearly along each boundary edge. Use the two P_1 triangles shown below.



$$\boldsymbol{n}_{\mathcal{F}} = \frac{(-1,1)}{\sqrt{2}}$$

All vertices lie on the boundary. The unique linear function matching the three nodal values on each triangle is

$$u_h^+(x_1,x_2) = x_1, \qquad u_h^-(x_1,x_2) = x_2.$$

Therefore,

$$\nabla u_h^+ = (1,0), \qquad \nabla u_h^- = (0,1).$$

Because $f = 0$ and these gradients are constant,

$$r_\pm = \nabla\cdot(\nabla u_h^\pm) = 0.$$

The boundary data are imposed exactly, so $e = 0$ on $\partial\Omega$.

Compute the facet term and the local indicators

Using the normal shown on the previous slide,

$$\begin{aligned} j_{\mathcal{F}} &= -\big(\nabla u_h^+ - \nabla u_h^-\big) \cdot \boldsymbol{n}_{\mathcal{F}} \\ &= -\big((1,0) - (0,1)\big) \cdot \frac{(-1,1)}{\sqrt{2}} = \sqrt{2}. \end{aligned}$$

The diagonal has length $h_{\mathcal{F}} = \sqrt{2}$, so

$$\|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} = \int_{\mathcal{F}} 2 \, \mathrm{d}s = 2\sqrt{2}, \qquad \eta^2 = h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})} = 4.$$

The shared-facet contribution is divided equally between its two neighboring elements:

$$\eta_+^2 = \eta_-^2 = 0 + \frac{1}{2}\,(4) = 2, \qquad \eta^2 = \eta_+^2 + \eta_-^2.$$

Since $r_\pm = 0$ and there are no Neumann facets, the residual formula reduces to $R(u_h; v) = \sqrt{2} \int_{\mathcal{F}} v \, \mathrm{d}s$. For example, $v = x_1(1-x_1)x_2(1-x_2)$ is zero on $\partial\Omega$ and positive inside $\mathcal{F}$, so $R(u_h; v) > 0$. Element terms alone miss this residual.

Reliability of the estimator

Combine the weighted residual bounds and obtain a global energy-error estimate

Combining the weighted residual bounds

With $w = e - \mathcal{Q}_h e$ and $\mathcal{F}_h^* = \mathcal{F}_h^{\text{int}} \cup \mathcal{F}_h^N$, start from

$$\begin{aligned}\|e\|_a^2 &= \sum_{e'} \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^*} \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \\ &\leq \sum_{e'} \left| \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \right| + \sum_{\mathcal{F} \in \mathcal{F}_h^*} \left| \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \right|.\end{aligned}$$

Apply the element and facet bounds to these absolute values:

$$\begin{aligned}\|e\|_a^2 \leq C_Q \Bigg[&\sum_{e'} h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, |e|_{H^1(\omega_{e'})} \\ &+ \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}}^{1/2} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \, |e|_{H^1(\omega_{\mathcal{F}})} \Bigg].\end{aligned}$$

The two lists in Cauchy–Schwarz

For nonnegative numbers A_i and B_i, Cauchy–Schwarz gives

$$\sum_i A_i B_i \le \left(\sum_i A_i^2 \right)^{1/2} \left(\sum_i B_i^2 \right)^{1/2}.$$

Use one combined list, with an entry for every element and every facet:

Entry	A_i: weighted residual	B_i: patch seminorm
Element e'	$h_{e'} \, \lVert r_{e'} \rVert_{L^2(\mathcal{K}_{e'})}$	$\lvert e \rvert_{H^1(\omega_{e'})}$
Facet $\mathcal{F}$	$h_{\mathcal{F}}^{1/2} \, \lVert j_{\mathcal{F}} \rVert_{L^2(\mathcal{F})}$	$\lvert e \rvert_{H^1(\omega_{\mathcal{F}})}$

The sum $\sum_i A_i B_i$ is exactly the bracket in the preceding bound.

Applying Cauchy–Schwarz to both residual sums

Substitute the entries in the table into $\|e\|_a^2 \le C_Q \sum_i A_i B_i$ and apply Cauchy–Schwarz:

$$\|e\|_a^2 \le C_Q \underbrace{\left(\sum_{e'} h_{e'}^2 \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}^2 + \sum_{\mathcal{F} \in \mathcal{F}_h^*} h_{\mathcal{F}} \, \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2 \right)^{1/2}}_{\eta} \times \left(\sum_{e'} |e|_{H^1(\omega_{e'})}^2 + \sum_{\mathcal{F} \in \mathcal{F}_h^*} |e|_{H^1(\omega_{\mathcal{F}})}^2 \right)^{1/2}.$$

The first factor is exactly the estimator already defined. The remaining task is to bound the sum of the patch seminorms.

Finite patch overlap gives a global error norm

Let $m_{e'}$ be the number of element and facet patches containing $\mathcal{K}_{e'}$. Shape regularity gives a mesh-independent bound $m_{e'} \leq N_{\mathrm{ov}}$. Expand each squared seminorm as an integral:

$$\begin{aligned}
&\sum_{e'} |e|^2_{H^1(\omega_{e'})} + \sum_{\mathcal{F}\in\mathcal{F}_h^*} |e|^2_{H^1(\omega_{\mathcal{F}})} \\
&= \sum_{e'} m_{e'} \int_{\mathcal{K}_{e'}} |\nabla e|^2 \, \mathrm{d}x \\
&\leq N_{\mathrm{ov}} \sum_{e'} \int_{\mathcal{K}_{e'}} |\nabla e|^2 \, \mathrm{d}x = N_{\mathrm{ov}} \, |e|^2_{H^1(\Omega)} \, .
\end{aligned}$$

Thus each element integral is counted at most N_{ov} times. Take the square root and substitute into the second Cauchy–Schwarz factor:

$$\|e\|_a^2 \leq C_Q \sqrt{N_{\mathrm{ov}}} \, \eta \, |e|_{H^1(\Omega)} \, .$$

Reliability of the residual estimator

Since $k(\boldsymbol{x}) \geq k_0 > 0$, the H^1 seminorm satisfies

$$|e|^2_{H^1(\Omega)} = \int_\Omega |\nabla e|^2 \, \mathrm{d}x \leq \frac{1}{k_0} \int_\Omega k|\nabla e|^2 \, \mathrm{d}x = \frac{1}{k_0} \|e\|^2_a .$$

In $\|e\|^2_a \leq C_Q \sqrt{N_{\mathrm{ov}}}\, \eta \, |e|_{H^1(\Omega)}$, replace the seminorm by $\|e\|_a / \sqrt{k_0}$:

$$\|e\|^2_a \leq C_Q \sqrt{\frac{N_{\mathrm{ov}}}{k_0}}\, \eta \, \|e\|_a .$$

If $e \neq 0$, divide by $\|e\|_a$. If $e = 0$, the bound is immediate:

$$\boxed{\|u - u_h\|_a \leq C_Q \sqrt{\frac{N_{\mathrm{ov}}}{k_0}}\, \eta.}$$

This is the reliability bound, with $C_{\mathrm{rel}} = C_Q \sqrt{N_{\mathrm{ov}}/k_0}$ independent of mesh size.

What reliability tells us

The result just proved is

$$\|u - u_h\|_a \leq C_{\mathrm{rel}}\eta.$$

On a shape-regular mesh family, C_{rel} stays fixed as the mesh is refined. Consequently,

$$\eta \longrightarrow 0 \quad \Longrightarrow \quad \|u - u_h\|_a \longrightarrow 0.$$

The computed η is *not itself* a guaranteed numerical upper bound: the theorem contains the constant C_{rel}, which we have not computed.
Likewise, η_e is a contribution to the estimator, not the exact error on $\mathcal{K}_e$. Reliability is a global statement. A separate estimate is needed to compare η_e with the error on its surrounding patch.

Returning to 1D: the nodal terms in the error bound

On an interval, $e \in H^1(0,1)$ has nodal values. We can choose $v_h = \mathcal{I}_h e \in \mathcal{V}_{h,0}$ and set $w = e - \mathcal{I}_h e$. At each node, $w(x_i) = e(x_i) - (\mathcal{I}_h e)(x_i) = 0$.
Start from the subtraction identity, then use the 1D residual formula:

$$\begin{aligned}
\|e\|_a^2 &= R(u_h; e) - \underbrace{R(u_h; \mathcal{I}_h e)}_{0} = R(u_h; w) \\
&= \sum_{e'=1}^{2} \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \underbrace{-\frac{1}{2} w(x_2) - \frac{1}{3} w(x_3)}_{0} \\
&\leq \sum_{e'=1}^{2} \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, \|w\|_{L^2(\mathcal{K}_{e'})} \, .
\end{aligned}$$

The last step uses Cauchy–Schwarz on each element. The nodal residuals are nonzero, but their products with $w(x_i)$ vanish.

The 1D interpolation bound and Cauchy–Schwarz

For linear nodal interpolation on an interval, use the estimate

$$\|v - \mathcal{I}_h v\|_{L^2(\mathcal{K}_{e'})} \le h_{e'} \, \|\mathrm{d}v/\mathrm{d}x\|_{L^2(\mathcal{K}_{e'})}, \qquad v \in H^1(0,1).$$

Use this estimate from Chapter 8 without proof. Set $v = e$, so $v - \mathcal{I}_h v = w$, then apply Cauchy–Schwarz to the sum:

$$\begin{aligned} \|e\|_a^2 &\le \sum_{e'=1}^{2} h_{e'} \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \, \|\mathrm{d}e/\mathrm{d}x\|_{L^2(\mathcal{K}_{e'})} \\ &\le \underbrace{\left(\sum_{e'=1}^{2} h_{e'}^2 \, \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}^2 \right)^{1/2}}_{\eta_{1\mathrm{D}}} \left(\sum_{e'=1}^{2} \|\mathrm{d}e/\mathrm{d}x\|_{L^2(\mathcal{K}_{e'})}^2 \right)^{1/2}. \end{aligned}$$

Since the elements partition $(0,1)$ and $k = 1$, the last factor is $\|\mathrm{d}e/\mathrm{d}x\|_{L^2(0,1)} = \|e\|_a$. Cancel $\|e\|_a$ when nonzero to obtain $\|e\|_a \le \eta_{1\mathrm{D}}$. For $e = 0$, this bound is immediate.

The actual element errors in the 1D example

The exact solution is $u(x) = x - x^2/2$: it satisfies $-u'' = 1$, $u(0) = 0$, and $u'(1) = 0$. Thus $u' = 1 - x$. Subtract the two computed slopes:

$$(u - u_h)' = \begin{cases} (1-x) - 5/6 = 1/6 - x, & x \in \mathcal{K}_1 = (0, 1/3), \\ (1-x) - 1/3 = 2/3 - x, & x \in \mathcal{K}_2 = (1/3, 1). \end{cases}$$

Since $k = 1$, integrate the square to obtain each local energy error:

$$\int_{\mathcal{K}_1} |u' - u_h'|^2 \, \mathrm{d}x = \int_0^{1/3} (x - 1/6)^2 \, \mathrm{d}x = \left[\frac{(x - 1/6)^3}{3} \right]_0^{1/3} = \frac{1}{324},$$

$$\int_{\mathcal{K}_2} |u' - u_h'|^2 \, \mathrm{d}x = \int_{1/3}^1 (x - 2/3)^2 \, \mathrm{d}x = \left[\frac{(x - 2/3)^3}{3} \right]_{1/3}^1 = \frac{8}{324}.$$

Element size in the same 1D example

Recall $h_1 = 1/3$, $h_2 = 2/3$, and $r_1 = r_2 = 1$. For each element, the squared contribution to $\eta_{1\mathrm{D}}$ is

$$h_e^2 \, \|r_e\|^2_{L^2(\mathcal{K}_e)} = h_e^2 \int_{\mathcal{K}_e} 1 \, \mathrm{d}x = h_e^2 h_e = h_e^3.$$

Compare these contributions with the squared errors just calculated:

Element	h_e	h_e^3	$\displaystyle\int_{\mathcal{K}_e} \lvert u' - u_h' \rvert^2 \, \mathrm{d}x$
$\mathcal{K}_1$	$1/3$	$1/27$	$1/324$
$\mathcal{K}_2$	$2/3$	$8/27$	$8/324$

Here $h_2 = 2h_1$, so $h_2^3/h_1^3 = 2^3 = 8$. The squared local errors also have ratio 8. The residuals are equal, but element size makes the longer element contribute more to both the estimator and the error.

Summary and next step

- $R(u_h; v) = a(u - u_h, v)$ connects the computable residual functional to the unknown error.
- Elementwise integration by parts separates balance defects inside elements from flux defects on interior and Neumann facets.
- Galerkin orthogonality gives $R(u_h; v) = R(u_h; v - \mathcal{Q}_h v)$. Local approximation estimates then supply the powers of the mesh size.
- The indicators η_e identify where the residual is concentrated, while reliability bounds the global energy error by $C_{\mathrm{rel}}\eta$.

Exit question. For P_1 elements with elementwise-constant k and $f = 0$, can η still be nonzero? Identify the residual term that remains.

Next lecture. Use the local indicators to mark elements, refine the mesh conformingly, and repeat the solve–estimate–mark–refine loop.

References for the construction and estimates

- J. Schöberl, *Numerical Methods for Partial Differential Equations*, §4.3, pp. 64–67. Definition 74 gives the patch-average construction. Theorems 75–76 give its local estimates and reliability. Online version: §27.1–27.2.
- D. N. Arnold, *Lecture Notes on Numerical Analysis of Partial Differential Equations*, Chapter 4, §9.1–9.4, pp. 67–73. Equations (4.16)–(4.18) give local estimates for a projection-based Clément operator. Equations (4.19)–(4.21) derive the residual bound.
- M. G. Larson and F. Bengzon, *The Finite Element Method: Theory, Implementation, and Practice*, §4.9.1, pp. 88–89, Theorem 4.7. See also §2.6.1, pp. 38–39, for the 1D argument.

The references develop the homogeneous Dirichlet case. Our proof retains the Neumann residual and uses the boundary-preserving estimates stated in Chapter 8 as Hypothesis (Q).